

Problem

Some modern power transmission systems now have major high-voltage DC transmission segments. There are a lot of good reasons for doing this but we will not go into them here. To go from the AC to DC, power electronics are used. We start with three-phase AC and then rectify it (using a full-wave rectifier). It was found that using a delta to wye and delta combination connected secondary would give us a much smaller ripple after the full-wave rectifier. How is this accomplished? Remember that these are real devices and are wound on common cores.

Hint: Use Figs. 13.47 and 13.49, and the fact that each coil of the wye connected secondary and each coil of the delta connected secondary are wound around the same core of each coil of the delta connected primary so the voltage of each of the corresponding coils are in phase. When the output leads of both secondaries are connected through full-wave rectifiers with the same load, you will see that the ripple is now greatly reduced. Please consult the instructor for more help if necessary.

Figure 13.47: Δ - Δ three-phase transformer connection.

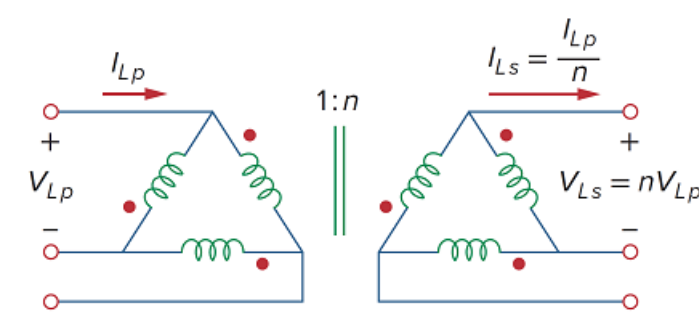
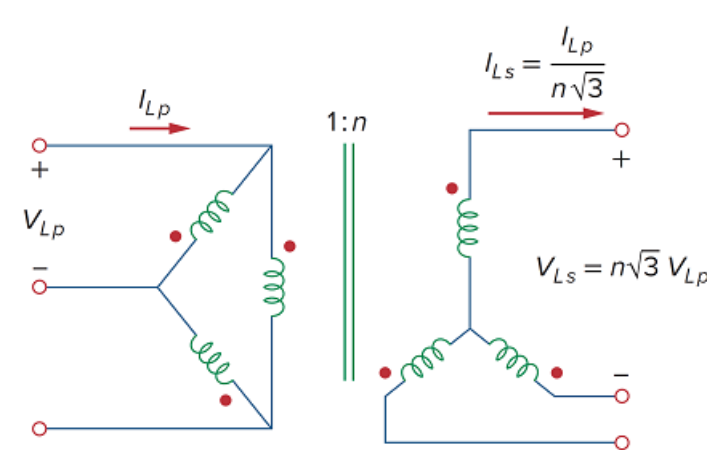


Figure 13.49: Δ -Y three-phase transformer connection.



Step-by-step solution

Step 1 of 1

Problem,

Some modern power transmission systems now have major, high voltage DC transmission segments. There are a lot of good reasons for doing this but we will not go into them here. To go from the AC to DC, power electronics are used. We start with three-phase AC and then rectify it (using a full-wave rectifier). It was found that using a delta to wye and delta combination connected secondary would give us a much smaller ripple after the full-wave rectifier. How is this accomplished? Remember that these are real devices and are wound on common cores. Hint, using Figures 13.47 and 13.49, and the fact that each coil of the wye connected secondary and each coil of the delta connected secondary are wound around the same core of each coil of the delta connected primary so the voltage of each of the corresponding coils are in phase. When the output leads of both secondaries are connected through full-wave rectifiers with the same load, you will see that the ripple is now greatly reduced. Please consult the instructor for more help if necessary.

This is a most interesting and very practical problem. The solution is actually quite easy, you are creating a second set of sine waves to send through the full-wave rectifier, 30° out of phase with the first set. We will look at this graphically in a minute. We begin by showing the transformer components.

The key to making this work is to wind the secondary coils with each phase of the primary. Thus, a-b is wound around the same core as A1-N1 and A2-B2. The next thing we need to do is to make sure the voltages come out equal. We need to work the number of turns of each secondary so that the peak of VA1–VB1 is equal to VA2–VB2. Now, let us look at some of the equations involved.

If we let $v_{ab}(t) = 100\sin(t)$ V, assume that we have an ideal transformer, and the turns ratios are such that we get $v_{A1-N1}(t) = 57.74\sin(t)$ V and $v_{A2-B2}(t) = 100\sin(t)$ V. Next, let us look at $v_{bc}(t) = 100\sin(t+120^\circ)$ V. This leads to $v_{B1-N1}(t) = 57.74\sin(t+120^\circ)$ V. We now need to determine $v_{A1-B1}(t)$.

$$v_{A1-B1}(t) = 57.74\sin(t) - 57.74\sin(t+120^\circ) = 100\sin(t-30^\circ) \text{ V.}$$

This then leads to the output per phase voltage being equal to $v_{out}(t) = [100\sin(t) + 100\sin(t-30^\circ)]$ V. We can do this for each phase and end up with the output being sent to the full-wave rectifier. This looks like $v_{out}(t) = [|100\sin(t)| + |100\sin(t-30^\circ)| + |100\sin(t+90^\circ)| + |100\sin(t-120^\circ)| + |100\sin(t-150^\circ)|]$ V. The end result will be more obvious if we look at plots of the rectified output.

In the plot below we see the normalized (1 corresponds to 100 volts) ripple with only one of the secondary sets of windings and then the plot with both. Clearly the ripple is greatly reduced!

